

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ac7c0d6192dc2c14e.jsonl`  
- SHA-256: `22e87209f28815abae8eaebcd15650af7dd8698071244ce654c6b2221ae57b16`  
- Source modified: `2026-06-16T03:28:30+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:27:26.424Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"The system detects hit moments primarily by finding direction changes in the shuttle trajectory (TrackNet's Peak Identification and Direction Identification Methods), then reconciles these with a separate swing-action detector through the Shot Refinement Algorithm to produce precise contact points. This is the exact mechanism missing from a pipeline that conflates 'shuttle moving' with 'rally in progress' ? trajectory direction-reversal/peak detection is the trajectory feature that marks a shot/contact event."

****Source:**** <https://www.mdpi.com/1424-8220/24/13/4372> (primary)

****Supporting quote:**** "TrackNet proposed the Direction Identification Method, which checks for sudden changes in the direction of the shuttlecock. ... The objective of the proposed SRA is to combine the results of both modules, thereby providing the precise identification of the hit moment."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:27:29.058Z)

I'll verify this claim by examining the source paper and checking whether the quote actually supports the specific mechanism described.

3. assistant (2026-06-16T03:27:31.804Z)

I'll fetch the actual paper to verify whether the quote supports the specific mechanism described in the claim.

4. user (2026-06-16T03:27:34.573Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. assistant (2026-06-16T03:27:36.567Z)

The MDPI page blocked direct access. Let me search for the paper and related information.

6. user (2026-06-16T03:27:45.899Z)

Web search results for query: "MDPI Sensors 2024 24 4372 badminton hit moment Shot Refinement Algorithm SRA TrackNet Direction Identification"

Links: [{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.mdpi.com/1424-8220/24/13/4372"}, {"title":"(PDF) Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.researchgate.net/publication/382039345_Enhancing_Badminton_Game_Analysis_An_Approach_to_Shot_Refinement_via_a_Fusion_of_Shuttlecock_Tracking_and_Hit_Detection_from_Monocular_Camera"}, {"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}, {"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"}, {"title":"Fine reconstruction of badminton swing dynamic trajectory assisted by event camera | Scientific Reports","url":"https://www.nature.com/articles/s41598-026-46443-8"}, {"title":"Use predictive analytics and big data technologies to enhance badminton game strategy development and performance indicators - Xin Feng, Xiang Liu, 2025","url":"https://journals.sagepub.com/doi/10.1177/14727978251324125"}]

I found the research paper you were looking for. Here's a summary of the key information:

Paper Details

The paper was published in Sensors 2024, Volume 24, Issue 13, Article 4372, on July 5, 2024. The authors are Yi-Hua Hsu, Chih-Chang Yu, and Hsu-Yung Cheng from Chung Yuan Christian University and National Central University in Taiwan.

Research Focus

The study proposes a novel method to extract shots in badminton games from a monocular camera, with TrackNet, a deep neural network designed for tracking small objects, used to extract the flight trajectory of the shuttlecock. The YOLOv7 model is used to identify whether the player is swinging.

Shot Refinement Algorithm (SRA)

Since both TrackNet and YOLOv7 may have detection misses and false detections, the study

proposes a shot refinement algorithm to obtain the correct hitting moment. The objective of the proposed SRA is to combine the results of both modules, thereby providing the precise identification of the hit moment.

Results

The proposed method achieves an accuracy of 89.7%, a recall rate of 91.3%, and an F1 rate of 90.5% in 69 matches, with 1582 rallies of Badminton World Federation match videos. This is a significant improvement compared to TrackNet alone, which yields 58.8% accuracy, 93.6% recall, and 72.3% F1 score.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-06-16T03:27:49.998Z)

Web search results for query: "TrackNet badminton "Peak Identification" "Direction Identification" shuttlecock hit detection direction change"

Links: [{"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.mdpi.com/1424-8220/24/13/4372"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}, {"title": "TrackNetV3: Enhancing Shuttlecock Tracking with Augmentations and Trajectory Rectification", "url": "https://dl.acm.org/doi/fullHtml/10.1145/3595916.3626370"}, {"title": "CoachAI: A Project for Microscopic Badminton Match Data Collection and Tactical Analysis", "url": "https://arxiv.org/pdf/1907.12888"}, {"title": "Track Net V3 | PDF", "url": "https://www.scribd.com/document/853585931/TrackNetV3"}, {"title": "TrackNetV3: Enhancing Shuttlecock Tracking with Augmentations and Trajectory Rectification | Proceedings of the 5th ACM International Conference on Multimedia in Asia", "url": "https://dl.acm.org/doi/10.1145/3595916.3626370"}, {"title": "Effectiveness of Advanced Tracking Models for Shuttlecock and Court Line Detection in Small-Scale Badminton Matches | SpringerLink", "url": "https://link.springer.com/chapter/10.1007/978-3-031-97578-3_71"}, {"title": "One-Shot Badminton Shuttle Detection for Mobile Robots", "url": "https://arxiv.org/pdf/2603.06691"}]

Based on the search results, here's information about TrackNet's badminton shuttlecock hit detection methods:

Peak Identification Method

After converting trajectories, the shuttlecock's trajectories form a wave-like pattern and hit moments primarily occur at the peak of this wave pattern. The frame of a hit can be identified by detecting those peaks.

Direction Identification Method

TrackNet proposed the Direction Identification Method, which checks for sudden changes in the direction of the shuttlecock. If there is a significant change in the direction of the shuttlecock, it is considered a hit that forces the shuttlecock to change its flight direction.

Why Both Methods Are Needed

Some trajectories do not exhibit obvious changes in y. To address this problem, TrackNet proposed the Direction Identification Method, which checks for sudden changes in the direction of the shuttlecock. This shows that the two methods complement each other?the Peak Identification Method works well for trajectories with obvious vertical changes, while the Direction Identification Method handles cases where the shuttlecock changes direction without obvious vertical movement.

The shuttlecock-tracking module includes the "Peak Identification Method" and "Direction Identification Method" to detect the sudden direction change of the shuttlecock.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:27:52.428Z)

The search confirms substantial detail. Let me fetch the open-access PMC version to verify the exact passages and the specific claim structure.

9. user (2026-06-16T03:28:00.891Z)

Badminton Analysis Methods: Peak Identification, Direction Identification, and SRA

(1) Peak Identification Method

"The Peak Identification Method converts the shuttlecock position from (x, y) image coordinates to (fi, y), where fi represents the frame number. After this conversion, the shuttlecock's trajectories will form a wave-like pattern and the hit moments primarily occur at the peak of this wave pattern."

(2) Direction Identification Method

"The Direction Identification Method checks for sudden changes in the direction of the shuttlecock. If there is a significant change in the direction of the shuttlecock, it is considered a hit that forces the shuttlecock to change its flight direction."

(3) Module Classification

Both methods belong to the **shuttlecock tracking module**. The document states: "Huang et al. [58] found that most of the hit action occurs at the relatively low point of the trajectory of the shuttlecock...Therefore, they proposed two algorithms to identify shots, including the 'Peak Identification Method' and the 'Direction Identification Method'." These are components of TrackNet's hit detection framework.

(4) Swing/Hit Detection Module

YOLOv7 is the model used. The document explains: "The model used in this study was YOLOv7...where a high confidence score in a specific frame implies that a player is performing a hit."

(5) Shot Refinement Algorithm (SRA)

SRA combines two modules: the shuttlecock tracking results (trajectory-based hit detection) and the YOLOv7 action detection results. Its objective is stated as follows:

"The objective of the proposed SRA is to combine the results of both modules, thereby providing the precise identification of the hit moment."

10. assistant (2026-06-16T03:28:12.305Z)

I now have all the verification needed. Let me assess the claim against the source.

The claim states the system "detects hit moments primarily by finding direction changes in the shuttle trajectory (TrackNet's Peak Identification and Direction Identification Methods), then reconciles these with a separate swing-action detector through the Shot Refinement Algorithm to produce precise contact points."

Verification against the paper:

- Peak Identification Method + Direction Identification Method: CONFIRMED ? both are trajectory-based methods (part of the shuttlecock tracking module, attributed to TrackNet/Huang et al.). Direction Identification checks for "sudden changes in the direction of the shuttlecock" = a hit.
- Separate swing-action detector: CONFIRMED ? YOLOv7 detects whether a player is swinging/performing a hit.
- Shot Refinement Algorithm reconciles both: CONFIRMED ? "combine the results of both modules... precise identification of the hit moment."

One nuance to scrutinize for the adversarial check: the claim says "primarily by finding direction changes." The paper actually uses BOTH Peak Identification AND Direction Identification (complementary). Saying "primarily... direction changes" slightly underweights Peak Identification, but the claim explicitly names BOTH methods in parentheses, so this is minor phrasing, not a misread. The core technical mechanism is accurately represented.

The interpretive bridge ("This is the exact mechanism missing from a pipeline that conflates 'shuttle moving' with 'rally in progress'") is a reasonable analytical inference, not a claim about what the paper says. The factual substrate ? direction-reversal/peak detection marks a shot/contact event ? is directly supported.

Source quality: peer-reviewed (Sensors/MDPI 2024), open-access, with reported metrics (89.7% accuracy, F1 90.5% vs TrackNet alone 72.3%). Current (2024). Primary source for the specific mechanism. License note: MDPI Sensors papers are CC-BY (the method/algorithm is described, not redistributed weights), so citing it is fine; the technique itself uses TrackNet (MIT) + YOLOv7 (whose weights are GPL/AGPL ? relevant if implemented, but that's an implementation-license concern, not a refutation of the descriptive claim).

The claim is well-supported, current, and from a sufficient primary source.

11. user (2026-06-16T03:28:23.574Z)

Structured output provided successfully

12. assistant (2026-06-16T03:28:30.591Z)

Verification complete. The claim is NOT refuted (refuted=false).

I verified the claim against the open-access primary source (Sensors 2024, 24(13):4372, PMC11244353) and confirmed every load-bearing element:

- Direction Identification Method and Peak Identification Method are both trajectory-based hit-detection methods (TrackNet/Huang et al. lineage), with Direction Identification checking for "sudden changes in the direction of the shuttlecock."
- A separate swing-action detector (YOLOv7) detects whether a player is performing a hit.
- The Shot Refinement Algorithm (SRA) combines both modules for "precise identification of the hit moment."

The claim is well-supported, current (2024), and from a peer-reviewed primary source with reported numbers. Two non-fatal nits flagged in the output: the "primarily direction changes" phrasing slightly underweights Peak Identification (but both are named), and the "exact mechanism missing" framing is the report-author's sound inference rather than a paper claim. I also flagged a license caveat for the project: the YOLOv7 swing detector is GPL/AGPL (incompatible with the MIT/Apache/BSD-only constraint if reused), while the trajectory direction-change idea via TrackNet (MIT) is clean.